PROG101 - Assignment 1

WasteTrack SL: Algorithm

Community Waste Collection Scheduling and Missed-Pickup Reporting

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2.1 Algorithm A: Handle a Missed-Pickup Report

1. Start.
2. Receive the resident's phone number and zone name (input).
3. Check whether the household is registered.
4. If it is not registered, show a registration prompt and go to step 17.
5. Look up the zone's scheduled collection day.
6. If today is not the collection day, show the next collection date and go to step 17.
7. Check whether the crew has marked the zone as collected today.
8. If it was collected, record the report as "Disputed", notify the supervisor to verify, and go to step 17.
9. If it was not collected, record the report as "Missed Pickup".
10. Count the missed-pickup reports for this zone in the last 7 days.
11. If the count is 3 or more, escalate to the supervisor.
12. Otherwise, assign the zone to the next available crew.
13. Send a confirmation message to the resident (output).
14. Wait until the crew marks the zone as collected.
15. Update the report status to "Resolved".
16. Notify the resident that the waste has been collected (output).
17. End.

2.2 Algorithm B: Send Daily Reminders

1. Start.
2. Set tomorrow equal to today plus one day.
3. Get the list of zones that are scheduled for collection tomorrow.
4. Set sentCount to 0.
5. For each zone in the list, get the households registered in that zone.
6. For each household, send an SMS reminder to its phone number.
7. Add 1 to sentCount after each message is sent.
8. Repeat steps 5 to 7 until all zones have been processed.
9. Display the total number of reminders sent (output).
10. End.

2.3 Algorithm C: Register a Household

1. Start.
2. Receive the resident's name, phone number and zone name (input).
3. Check that the phone number has at least 8 digits and contains only digits.
4. If the phone number is invalid, display "Invalid phone number" and go to step 9.
5. Check that the zone exists.
6. If the zone does not exist, display "Zone not found" and go to step 9.
7. Generate a unique household ID and save the household details.
8. Send an SMS to the resident with the household ID (output).
9. End.

2.4 Algorithm D: Generate the Weekly Report

1. Start.
2. Get the list of all zones.

3. For each zone, count the scheduled collections and the missed reports for the last 7 days.
4. If scheduled collections is greater than 0, calculate pickup rate as ((scheduled - missed) / scheduled) x 100. Otherwise, set pickup rate to 0.
5. If pickup rate is 90 or more, set the rating to "Good".
6. Otherwise, if pickup rate is 60 or more and less than 90, set the rating to "Needs Improvement".
7. Otherwise, set the rating to "Critical".
8. Display the zone, scheduled collections, missed reports, pickup rate and rating (output).
9. Repeat steps 3 to 8 until all zones are done.
10. End.